

Multiple Choice (10 marks)

1. What is the value of this python expression: $4 + 3 \% 2$?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
2. In Python 3, what is the output of `print tuple[0]` if `tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )`?
 - (a) ('abcd', 786 , 2.23, 'john', 70.2)
 - (b) abcd
 - (c) Error
 - (d) None of the above.
3. In Python 3, which of the following function convert a string to an int in python?
 - (a) `int(x [,base])`
 - (b) `long(x [,base] )`
 - (c) `float(x)`
 - (d) `str(x)`
4. Which of the following is the hexadecimal representation of the decimal number 231_{10} ?
 - (a) 17_{16}
 - (b) $E4_{16}$
 - (c) $E7_{16}$
 - (d) $F4_{16}$
5. Which types of functions grow the slowest?
 - (a) $O(N^{1/2})$
 - (b) $O(N^{1/4})$
 - (c) $O(N^{1/N})$
 - (d) $O(N)$

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the function `max(list)` returns the item with the highest value from the given list.
7. True or False: The user's public key used for encryption poses the greatest risk to personal privacy among online data.
8. True or False: The value of the Python expression `1 + 3 % 3` is 1.
9. True or False: In Python 3, the `join()` function checks if all characters in a string are uppercase.
10. True or False: In Python 3, the operator `'**'` is used to perform exponential (power) calculations on operands.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate arc length of an angle.
12. Write a python function to left rotate the string.

End of Paper